

Optimal Timing of Acustimulation for Antiemetic Prophylaxis as an Adjunct to Ondansetron in Patients Undergoing Plastic Surgery

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We designed this study to evaluate the antiemetic efficacy of transcutaneous electrical acupoint stimulation in combination with ondansetron when applied before, after, or both before and after plastic surgery. A randomized, double-blind, sham-controlled study design was used to compare three prophylactic acustimulation treatment schedules: preoperative—an active device was applied for 30 min before and a sham device for 72 h after surgery; postoperative—a sham device was applied for 30 min before and an active device for 72 h after surgery; and perioperative—an active device was applied for 30 min before and 72 h after surgery ($n = 35$ per group). All patients received a standardized general anesthetic, and ondansetron 4 mg IV was administered at the end of surgery. The incidence of vomiting/retching and the need for rescue antiemetics were determined at specific time intervals for up to 72 h after surgery. Nausea scores were recorded with an 11-point verbal rating scale. Other outcome variables assessed included discharge times (for outpatients), resumption of normal activities of daily living, complete antiemetic response rate, and patient satisfaction with antiemetic therapy and quality of recovery. Perioperative

use of the ReliefBand® significantly increased complete responses (68%) compared with use of the device before surgery only (43%). Median postoperative nausea scores were significantly reduced in the peri- and postoperative (versus preoperative) treatment groups. Finally, patient satisfaction with the quality of recovery (83 ± 16 and 85 ± 13 vs 72 ± 18) and antiemetic management (96 ± 9 and 94 ± 10 vs 86 ± 13) on an arbitrary scale from 0 = worst to 100 = best was significantly higher in the groups receiving peri- or postoperative (versus preoperative) acustimulation therapy. For patients discharged on the day of surgery, the time to home readiness was significantly reduced (114 ± 41 min versus 164 ± 50 min; $P < 0.05$) when acustimulation was administered perioperatively (versus preoperatively). In conclusion, acustimulation with the ReliefBand® was most effective in reducing postoperative nausea and vomiting and improving patients' satisfaction with their antiemetic therapy when it was administered after surgery.

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There is still controversy concerning the best approach to managing postoperative nausea and vomiting (PONV) (1). Clinical studies suggest that the use of a combination of prophylactic antiemetic therapies improves patients' satisfaction with

their quality of recovery and facilitates resumption of normal activities (2–4). Although small-dose droperidol is a highly cost-effective antiemetic for routine prophylaxis (4–6), concerns regarding the side effects associated with larger doses (e.g., dysphoria, restlessness, and cardiac arrhythmias) (7,8) have limited its clinical use (9). As a result, ondansetron has become increasingly popular for routine antiemetic prophylaxis and appears to be most effective when administered at the end of surgery (10,11).

An earlier study demonstrated that the combination of ondansetron and acustimulation was more effective than ondansetron alone in preventing PONV (3). Although a meta-analysis by Lee and Done (12) confirmed that acustimulation was effective in preventing

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PONV, the optimal use of this nonpharmacologic technique has not been studied (13). In a preliminary multicenter study, use of transcutaneous electrical acustimulation reduced nausea but not vomiting when administered as the sole prophylactic therapy after laparoscopic surgery (14). However, more recent studies have reported a reduction in both nausea and emesis (3,15). Although it was suggested that preoperative versus postoperative acupoint stimulation made no difference in the incidence of vomiting within the first 6 h after major gynecologic surgery (16), this study was terminated prematurely.

Therefore, we designed a sham-controlled study to test the hypothesis that the timing of acustimulation with the ReliefBand® (Woodside Biomedical Systems, Inc., Carlsbad, CA) during the perioperative period influenced its efficacy when used as part of a multimodal regimen for antiemetic prophylaxis. The primary objective was to compare stimulation with the ReliefBand® before and/or after major plastic surgery procedures with respect to its effect in preventing PONV. Secondary objectives were to assess the effect of acustimulation therapy on the need for rescue antiemetic drugs and resumption of normal activities of daily living.

Methods

After we obtained IRB approval and written informed consent, 105 adults scheduled for elective plastic surgery (e.g., abdominoplasty, breast reductions, or facial cosmetic procedures) under general anesthesia were enrolled in this clinical study. Patients were randomly assigned to one of three acustimulation treatment groups by using a computer-generated random number table: 1) the preoperative group received an active ReliefBand® for 30 min before entering the operating room (OR) and a sham ReliefBand® for up to 72 h after surgery; 2) the postoperative group received a sham device for 30 min before surgery and an active ReliefBand® for up to 72 h after surgery; and 3) the perioperative group received an active ReliefBand® for 30 min before surgery and for up to 72 h after surgery. One patient in the perioperative group did not complete the entire study.

Patients who had taken any antiemetic medication within 24 h before the operation, were pregnant or menstruating, had a permanent cardiac pacemaker, had previous experience with acustimulation therapy, or had experienced nausea, vomiting, or retching within 24 h before surgery were excluded. Any history of PONV or motion sickness or active smoking was noted. In the day-surgery unit (DSU), patients completed a preoperative verbal rating scale (VRS) for nausea, with 0 = none to 10 = worst imaginable. An active (Groups 1 and 3) or sham (Group 2) ReliefBand® device was applied at the P₆ acupoint of the

nondominant arm for 30 min before the patient was transferred from the DSU to the OR. Patients were instructed in the correct placement of the device and told to apply it snugly around the wrist to avoid inadvertent displacement. Although the ReliefBand® devices used in all three groups were identical in appearance, the sham devices were electronically altered by the manufacturer to allow them to simulate the active unit without actually delivering electrical current to the metal electrodes in contact with the surface of the skin.

Before either the active or sham device was applied, the P₆ placement area on the plantar aspect of the wrist was carefully cleaned with an alcohol swab, followed by the application of a hypoallergenic conductivity gel. The controller setting on the device was initially turned to Setting 3 unless the patient complained of discomfort at the P₆ acupoint, in which case it was reduced to Setting 2. To minimize bias resulting from the presence or absence of the electrical stimulation, patients were told that the acustimulation device produces a tingling sensation that they "might or might not feel." The device was removed when the patient entered the OR.

All patients received a standardized general anesthetic technique consisting of propofol 1.5–2.5 mg/kg and sufentanil 0.3 µg/kg IV for induction, rocuronium 0.6 mg/kg IV for tracheal intubation, and sevoflurane 0.75%–3% end-tidal concentration in an air/oxygen mixture in combination with a sufentanil infusion ($0.1\text{--}0.2\text{ }\mu\text{g}\cdot\text{kg}^{-1}\cdot\text{h}^{-1}$) and rocuronium $0.075\text{--}0.15\text{ mg}\cdot\text{kg}^{-1}\cdot\text{h}^{-1}$ for maintenance of anesthesia. Residual neuromuscular blockade at the end of surgery was reversed with a combination of neostigmine 40–60 µg/kg and glycopyrrolate 5–10 µg/kg IV. All patients also received antiemetic prophylaxis with ondansetron 4 mg IV at the end of surgery. Upon patients' arrival in the postanesthesia care unit (PACU), a nurse specifically trained in the proper positioning of the ReliefBand® placed an active (Groups 2 and 3) or sham (Group 1) device at the P₆ acupoint of the nondominant arm at the same setting that was used before surgery. All patients were instructed to wear the acustimulation device for 72 h after surgery except when bathing and sleeping. It was recommended that the device be activated every 2–4 h during the awake hours and before consumption of opioid-containing analgesic medication. An antiemetic rescue drug (metoclopramide 10 mg IV) was administered only if the patient complained of persistent nausea (with a nausea VRS >3) lasting longer than 15 min or experienced repeated episodes of vomiting or retching.

The type and duration of the plastic surgery procedures, as well as the length of stay in the PACU and DSU, were recorded. All inpatients were started on patient-controlled analgesia with morphine (1-mg IV boluses) before discharge from the PACU. Postoperative nausea was assessed by a blinded observer with

the 11-point VRS upon admission to the PACU, at 15 and 30 min after the device was reapplied, and subsequently at 30-min intervals until the patient was discharged from the PACU. VRS nausea scores >3 on 2 consecutive assessments were considered to indicate clinically significant nausea. All episodes of vomiting and retching, as well as the need for rescue antiemetic medications, were recorded in a patient diary during the 72-h postoperative follow-up period. Complete responses were defined as no vomiting or retching (i.e., emesis) or need for a rescue antiemetic to treat persistent nausea (VRS >3) within the first 24 h after surgery.

For the outpatients ($n = 30$), home discharge criteria (home readiness) required that the patient be fully awake and oriented, have stable vital signs while standing, be able to walk without assistance, and not be experiencing any active side effects (e.g., pain, dizziness, or emetic symptoms) (5,10). Postoperative pain management consisted of oral opioid-containing analgesics. Follow-up visits to the postsurgical ward and/or telephone calls were performed at 24- and 72-h intervals after surgery by an investigator (who was unaware of the treatment group to which the patient was assigned) to inquire about any emetic symptoms and to evaluate the patient's satisfaction with the quality of recovery and antiemetic management by using a VRS, with 0 = poor recovery to 100 = excellent recovery and 0 = very dissatisfied to 100 = highly satisfied, respectively. At the time of the 24- and 72-h follow-up evaluations, patients were queried about their continued use of the acustimulation device, as well as their resumption of a normal diet, physical activities (e.g., ambulating and lifting), and a regular sleep pattern.

Assuming that 50% of these patients would develop nausea, vomiting, or both after plastic surgery under general anesthesia with ondansetron (4 mg IV) alone for antiemetic prophylaxis (3), a minimum sample size of 30 patients was determined by an *a priori* power analysis to provide an 80% power of detecting an absolute difference of 25% between treatment groups (i.e., a reduction from 50% to 25%) with an α of 0.05. Nonparametric analysis was used to compare nausea VRS scores, number of emetic episodes, and antiemetic use among the three study groups. The χ^2 test was used to compare the proportion of patients complaining of nausea (i.e., consecutive VRS nausea scores >3), having one or more emetic episodes, and receiving rescue antiemetic medication. In cases in which expected frequencies were small, Fisher's exact test was used. The times to the first emetic event and to rescue medication were analyzed with log-rank test statistics. The times to when 25% of the patients in each group were judged to have failed the prophylactic antiemetic therapy (i.e., required a rescue antiemetic drug for persistent nausea or repeated emesis) were determined by the Kaplan-Meier method. Data

analysis was performed with NCSS Version 6.0 (NCSS, Kaysville, UT). Data are presented as means $\pm$ SD, medians (interquartile ranges), numbers, or percentages, with P values <0.05 considered statistically significant.

Results

The three groups were not significantly different with respect to demographic characteristics (and types of surgery procedures), preexisting risk factors for developing PONV, preoperative nausea scores, and intraoperative dosages of propofol, sevoflurane, and sufentanil (Table 1). Postoperative opioid use was also similar among the three treatment groups in the PACU, on the postsurgical ward, and during the postdischarge period (Table 1). During the predischARGE period (<4 h), the need for rescue antiemetics was significantly reduced in the perioperative (versus preoperative) group (Table 2). In addition, the median VRS nausea scores were significantly lower in both the postoperative and perioperative (versus preoperative) groups during the predischARGE period and at the 24-h follow-up (Table 2). The overall incidences of PONV within 24 h after surgery were 57%, 46%, and 32% in the pre-, and post-, and perioperative groups, respectively.

The complete response rate was significantly higher in the perioperative (versus preoperative) group during the 4 to 24 h after surgery (Table 2). The percentages of patients who were able to resume a normal diet and sleep pattern on the first postoperative day did not differ significantly among the three groups. However, more patients in the perioperative group were able to resume physical activities at 24 h after surgery (21%, versus 3% in the preoperative group). Of note, a larger proportion of the patients undergoing plastic surgery procedures in the perioperative group were discharged home on the day of surgery (41%, versus 20% in the preoperative group). None of the patients were readmitted to the hospital because of intractable PONV. In the subset of patients discharged home on the day of surgery, the average length of stay in the DSU was significantly reduced in the perioperative (versus preoperative) group (114 ± 41 min versus 164 ± 50 min, respectively) (Table 2). Finally, the postoperative and perioperative groups had improved patient satisfaction with their quality of recovery and antiemetic therapy compared with the preoperative group (Table 2).

The Kaplan-Meier estimates (Fig. 1) suggested that the median time intervals for 25% of the patients to develop symptoms requiring antiemetic rescue therapy were 30, >240 , and >240 min in Groups 1, 2, and 3, respectively. No local (cutaneous) side effects were reported at the acustimulation site by any patient in the three treatment groups during the 72-h study period.

Table 1. Demographic Characteristics, Preoperative Nausea Scores, and Anesthesia and Surgery Times in the Three Acustimulation Treatment Groups

Variable	Preoperative (n = 35)	Postoperative (n = 35)	Perioperative (n = 34)
Age (yr)	40 ± 13	41 ± 13	41 ± 15
Sex (F/M) (n)	33/2	32/3	32/2
ASA status (I/II/III) (n)	17/16/2	13/19/3	14/17/3
Height (cm)	164 ± 6	165 ± 8	166 ± 8
Weight (kg)	67 ± 13	71 ± 15	69 ± 16
H/O PONV, n (%)	16 (46)	17 (49)	12 (34)
H/O motion sickness, n (%)	11 (31)	8 (23)	9 (26)
H/O postoperative dizziness, n (%)	5 (14)	5 (14)	4 (12)
Active smoker, n (%)	1 (3)	2 (6)	3 (9)
Type of plastic surgery (n)			
Head and neck	6	6	6
Breast reduction and augmentation	23	21	21
Abdominal and extremities	6	8	7
Anesthesia time (min)	204 ± 100	186 ± 98	190 ± 105
Propofol (mg)	142 ± 37	143 ± 39	148 ± 31
Sufentanil (μg)	61 ± 31	58 ± 32	57 ± 37
Sevoflurane (ET%)	1.8 ± 0.4	1.7 ± 0.4	1.8 ± 0.3
Surgery time (min)	172 ± 99	151 ± 89	167 ± 103
PACU fentanyl (μg)	126 ± 77	142 ± 79	118 ± 78
PCA morphine (mg)	9 ± 6	7 ± 7	11 ± 9
Oral opioid analgesics (n)	2.1 ± 1	1.5 ± 0.8	1.5 ± 0.5

Values are mean ± SD, unless otherwise noted.

There were no significant differences among the three groups.

H/O = history of; PACU = postanesthesia care unit; ET = end-tidal concentration; PCA = patient-controlled analgesia; PONV = postoperative nausea and vomiting.

Table 2. Recovery Times and Incidences of Postoperative Nausea, Vomiting, and Need for Rescue Antiemetic Medication in the Three Prophylactic Acustimulation Treatment Groups

Variable	Preoperative (n = 35)	Postoperative (n = 35)	Perioperative (n = 34)
Recovery times			
PACU, min (mean ± SD)	94 ± 40	85 ± 44	89 ± 48
Required overnight admission, n (%)	28 (80)	26 (74)	20 (69)
DSU stay, n (min) (mean ± SD)	7 (164 ± 50)	9 (124 ± 57)	14 (114 ± 41)*
During predischage period (<4 h)			
VRS nausea score, median (range)	2 (0-7)	0 (0-4)*	0 (0-3)*
Consecutive VRS nausea scores > 3, n (%)	10 (29)	6 (17)	3 (9)
Vomiting/retching, n (%)	7 (20)	5 (14)	4 (12)
Antiemetic rescue, n (%)	14 (40)	7 (20)	6 (18)*
Time to first emetic symptom, n (min) (mean ± SD)	20 (40 ± 66)	16 (58 ± 27)	11 (67 ± 34)
At 4-24-h follow-up evaluation			
Complete response, n (%)	15 (43)	19 (54)	23 (68)*
VRS nausea score, median (range)	4 (0-8)	0 (0-5)*	0 (0-4)*
Antiemetic rescue, n (%)	16 (46)	10 (29)	9 (26)
Resume normal diet, n (%)	20 (57)	25 (71)	24 (71)
Resume sleep pattern, n (%)	12 (34)	19 (54)	18 (53)
Resume physical activities, n (%)	1 (3)	2 (6)	7 (21)*
Quality of recovery (0-100) (mean ± SD)	72 ± 18	85 ± 13*	83 ± 16*
At 24-72-h follow-up evaluation			
VRS nausea score, median (range)	0 (0-3)	0 (0-1)	0 (0-1)
Vomiting, n (%)	0 (0)	0 (0)	0 (0)
Resume normal diet, n (%)	24 (69)	27 (77)	26 (76)
Resume sleep pattern, n (%)	18 (51)	24 (69)	22 (65)
Resume normal activities, n (%)	8 (23)	11 (31)	13 (38)
Satisfaction with antiemetic therapy (0-100) (mean ± SD)	86 ± 13	94 ± 10*	96 ± 9*

PACU = postanesthesia care unit; DSU = day-surgery unit; VRS = verbal rating scale (0 = none to 10 = maximal); PONV = postoperative nausea and vomiting.

* $P < 0.05$ versus preoperative group.

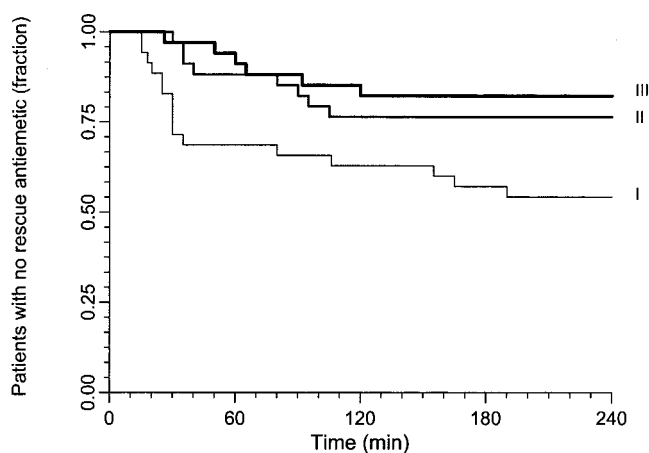


Figure 1. Kaplan-Meier estimates demonstrating the time to rescue with an antiemetic drug for patient complaints of either persistent nausea or vomiting after arrival in the postanesthesia care unit. Group 1 (Curve I) = preoperative stimulation; Group 2 (Curve II) = postoperative stimulation; Group 3 (Curve III) = perioperative stimulation. Compared with preoperative stimulation (I), perioperative stimulation (III) was significantly different at all time points after 20 min ($P < 0.05$).

Discussion

The clinical benefits of routine antiemetic prophylaxis for high-risk surgical patients have been well documented in the anesthesia literature (1,2,5,10,11,17,18). These benefits were not limited to cost savings for treatment of emetic episodes but also included improved patient satisfaction compared with simply treating established symptoms (10,11). Although multimodal antiemetic regimens involving up to three antiemetic drugs are justified in patients at high risk of developing PONV (2,17,18), the possibility of adverse drug interactions increases as a function of the number of drugs administered. This study examined the effect of timing of acustimulation when administered in combination with ondansetron for preventing postoperative emetic symptoms and improving patient outcomes. The enhanced antiemetic efficacy of acustimulation when administered in the postoperative (versus preoperative) period suggests that this nonpharmacologic therapy has little preemptive antiemetic effect. This observation was also confirmed by the fact that the efficacy of perioperative stimulation was not significantly better than postoperative stimulation alone.

Previous studies have demonstrated the antiemetic effectiveness of stimulation at the P_6 acupuncture point compared with a standard antiemetic drug (e.g., metoclopramide, droperidol, or ondansetron) (3,15,19–22). Although all three treatment groups probably received some benefit from the presence of acupressure at the P_6 acupoint, electrical stimulation clearly produced additional benefit when administered in the postoperative period. Interestingly, Kotani et al. (23) reported, in a sham-controlled study, that perioperative intradermal (nonelectrical) acupuncture

stimulation in a paravertebral distribution also reduced nausea and vomiting after major abdominal surgery. However, these authors did not evaluate the relative benefits of preoperative versus postoperative treatments on patient outcomes, and the apparent antiemetic action may have been related to an opioid-sparing effect. The study by Yentis and Vashisht (16) examined the antiemetic effect of acupuncture stimulation when administered before, during, or after surgery. Given the brief duration of the treatment sessions (five minutes), it is hardly surprising that the authors found no differences in the incidences of vomiting in the first six hours after major lower abdominal surgery. Their failure to examine the effect of the treatments on postoperative nausea may also have contributed to their negative findings. We are unaware of any other clinical studies that have systematically examined the antiemetic and antinausea efficacy of preoperative versus postoperative acustimulation.

Another controversy in this field of research relates to the relative effectiveness of acustimulation on nausea and vomiting (or emesis). In a preliminary multicenter study involving use of the ReliefBand[®] during the postoperative period, Zarate et al. (14) found a significant antinausea effect but failed to demonstrate a significant decrease in the incidence of emesis. These findings were confirmed in a more recent study by Rusy et al. (24) involving the use of electroacupuncture in children. However, a recent sham-controlled study by Gan et al. (15) found a reduction in both nausea and vomiting when acustimulation was administered with a Hans transcutaneous electrical stimulating device in the postoperative period. The ReliefBand[®] acustimulation device appears to be more effective for preventing than for treating PONV when used in combination with an antiemetic drug (3,25). The current study also confirms that the primary benefit of electroacustimulation was in reducing nausea and in improving patient satisfaction during the postoperative period.

Methodologic problems are common with clinical studies involving the use of nonpharmacologic antiemetic therapies. As reported in previous studies (3,25), patients receiving the active ReliefBand[®] devices after surgery are more likely to be able to detect the tingling sensation produced at the P_6 acupoint, and therefore patient bias may have contributed to the greater antiemetic efficacy of acustimulation in the groups receiving postoperative (versus preoperative) stimulation. Another possible deficiency in the study design related to the fact that there was no perioperative sham group. Because the efficacy of the ReliefBand[®] has been demonstrated in earlier sham-controlled studies involving similar high-risk patient populations (3,14), it was not considered necessary to include this group in this study. Although acupressure may contribute to the antiemetic efficacy of the

ReliefBand[®] device, this effect would have minimized the differences among the three treatment groups. Because differences in postoperative opioid analgesic use can also contribute to differences in PONV, the pain management regimen was standardized for all patients. A final criticism of our study design related to the mix of plastic surgery procedures included in this study. Although the distribution of cases was similar in all three treatment groups (Table 1), the ability of more patients in the perioperative group to resume physical activities at 24 hours after surgery may reflect the fact that more patients in this treatment group were discharged home on the day of surgery.

As a result of the traditionally frequent incidence of PONV in this plastic surgery population, ondansetron (4 mg IV) was administered to all patients (1,12). Ethical concerns raised by the surgical and anesthesia staff prevented us from including a study group receiving no antiemetic prophylaxis. Increasing interest in the use of acustimulation techniques is related in part to the recent controversy regarding the use of small-dose droperidol for antiemetic prophylaxis (6,8,9). Analogous to earlier P₆ acupoint stimulation studies in children (19,20,24) and adults (3,15,21,22), this clinical investigation confirms that the ReliefBand[®] is an effective adjunct to antiemetic drugs for preventing symptoms related to PONV when administered after surgery. Although the ReliefBand[®] device has been reported to enhance the antiemetic efficacy of ondansetron when administered in the postoperative period (3), 18%–20% of these high-risk patients still required a rescue antiemetic to control their emetic symptoms after surgery. These data would suggest that the addition of a second antiemetic drug (e.g., droperidol or dexamethasone) is probably indicated to minimize PONV after these highly emetogenic surgical procedures (26).

In conclusion, use of the ReliefBand[®] acustimulation device after plastic surgery appeared to enhance the antiemetic efficacy of ondansetron. However, the device is significantly less effective when applied only in the preoperative period.

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